

Data Provenance & Authenticity Verification

AI-Native Self-Healing O-RAN Network using a Digital Twin — Samsung PRISM worklet

This memo records an independent forensic check of every dataset used to produce the project's reported results, carried out on 17 August 2026 against the files as they exist on disk. Its purpose is to establish which data is **real measured network traffic**, which is **declared synthetic**, and what evidence supports each classification. All checks are reproducible with the commands in Section 6.

1 Summary verdict

Dataset	Classification	Basis
TIMESAFE captures data/external/timesafe_multi_raw/ data/external/timesafe_sessions/	REAL — third-party measured	Genuine libpcap captures from a physical 5G fronthaul testbed at Northeastern University, published with a peer-reviewed paper. Switch hardware identified from LLDP inside the capture itself.
Live netem runs results/tier2/netem/ results/live/	REAL — self-measured	Nanosecond-resolution captures of two genuine ptp4l daemons over a Linux virtual link. Message rates and clock identities match linuxptp defaults exactly.
Upstream repository data/external/s-plane_security_repo/	REAL — third-party	Verified git clone of a public research repository, with intact clone log and commit hash.
Simulator output (generated at runtime)	SYNTHETIC — declared	Physics-based generator. Produces no files stored in data/; created on demand during a run. Must never be presented as measured data.

Conclusion. Every dataset stored on disk under data/ is real measured traffic obtained from external published research or from the team's own instrumented runs. No fabricated, randomly generated, or placeholder data was found in any input used for the reported real-data results. One inventory error was found and is documented in Section 5.

2 TIMESAFE captures — evidence of authenticity

2.1 Published external source

Field	Verified value
Git remote	https://github.com/genesys-neu/s-plane_security.git
Commit	77a47ccb87a8170f1067f7783b12c9ed742ceb6f
Clone record	Present in .git/logs/HEAD, single clone entry, no local commits
Publication	TIMESAFE, <i>ACM Transactions on Privacy and Security</i> 28(5), 2025 — DOI 10.1145/3775060; preprint arXiv:2412.13049
Originating lab	GENESYS / Institute for the Wireless Internet of Things, Northeastern University

The repository is an unmodified clone: the working tree contains no local commits, so the captures are byte-for-byte as published by the original authors.

2.2 Capture-file forensics

Capture	Packets	Capture window (UTC)	Duration	Snaplen
announce_session_1.pcap	48,212	2024-10-06 04:09:07 → 04:20:11	664.6 s	262144
announce_session_2.pcap	48,212	2024-10-06 04:09:07 → 04:20:11	664.6 s	262144
announce_session_3.pcap	6,256	2024-10-08 04:44:06 → 04:45:36	90.1 s	8192

sync_followup_session.pcap	8,640	2024-10-08 04:45:52 → 04:47:10	77.7 s	8192
sync_singlestep_session.pcap	9,312	2024-10-08 04:47:20 → 04:48:51	91.1 s	8192

All five are standard libpcap v2.4, linktype 1 (Ethernet), with zero trailing bytes — that is, structurally complete files with no truncation or hand-editing. The two distinct snaplen values correspond to two separate capture sessions on two different days, consistent with independently configured tcpdump runs.

2.3 Physical testbed hardware, identified from inside the capture

The captures contain incidental LLDP (Link Layer Discovery Protocol) frames broadcast by the switch that carried the traffic. These are not part of the experiment and would not exist in a synthesised file. Decoding them yields:

```
SysName      = Xmili-ToR
SysDescr     = Dell SmartFabric OS10 Enterprise – OS Version 10.5.5.5 – System Type: S5232F-ON
PortDescr    = devkit02-FH
```

The Dell S5232F-ON is a commercially available 32-port 100 GbE data-centre switch. The port description devkit02-FH denotes the fronthaul port of development kit 02, and Xmili-ToR follows a standard top-of-rack naming convention. A fabricated dataset would not carry a coherent vendor operating-system banner with a valid release number.

2.4 Traffic-level indicators consistent only with a live network

- **Incidental infrastructure traffic.** Alongside PTP (ethertype 0x88f7) the captures contain Cisco PVST+ spanning-tree frames (01:00:0c:cc:cc:cd), LLDP (0x88cc), IPv6 neighbour discovery (33:33:ff:00:00:00), VLAN-tagged frames (0x8100) and broadcasts. Synthetic generators emit only the protocol under study.
- **Correct PTP multicast destination.** 01:1b:19:00:00:00, the IEEE 1588 standard address.
- **Telecom PTP profile.** All frames are PTPv2 with domainNumber = 24, the ITU-T G.8275.1 telecom-profile domain used for fronthaul timing — not the default domain 0.
- **Internally consistent protocol exchange.** In session 3: Delay_Req = 428 and Delay_Resp = 428, exactly matched request/response pairs, alongside Sync 2,461 / Follow_Up 1,019 / Announce 1,688.
- **Natural timing jitter.** 22,854 distinct inter-arrival values across 48,211 intervals. Generated data characteristically shows few repeated intervals.
- **Multiple real endpoints.** Eight distinct source MAC addresses with globally-administered vendor OUIs.

2.5 The analysis CSVs are provably derived from the captures

Each session's attack and benign CSV row counts were compared against the number of PTP frames in the corresponding capture. If the CSVs had been authored independently of the captures, these would not agree.

Session	Attack rows	Benign rows	Total	PTP frames in pcap	Match
announce_session_1	46,987	11	46,998	46,998	exact
announce_session_2	46,992	6	46,998	46,998	exact
announce_session_3	4,538	1,485	6,023	6,024	-1
sync_followup_session	7,424	1,016	8,440	8,440	exact
sync_singlestep_session	7,370	1,707	9,077	9,078	-1

3 Live netem runs — evidence of authenticity

These are the team's own measurements, so they were checked to the same standard.

- **Nanosecond-resolution libpcap** (magic 4d3cb2a1), which only a kernel/hardware timestamping capture path produces.

- **Capture dates 2026-07-27 and 2026-08-07**, the latter matching the repository's known-good-2026-08-07 restore tag.
- **Clock identities are correctly derived EUI-64 values.** c29a38fffeb34e68 is MAC c2:9a:38:b3:4e:68 with ff:fe inserted at the midpoint — exactly how linuxptp constructs a clockIdentity from an interface address.
- **Locally-administered address bit is set** on both endpoints, the expected signature of Linux veth virtual interfaces used by the netem harness — consistent with the documented method.
- **Message rates match ptp4l defaults.** Over 83.7 s: Sync 668 ($\approx 8/s$, logSyncInterval -3) and Announce 42 ($\approx 0.5/s$, logAnnounceInterval 1). Sync = Follow_Up = 668 and Delay_Req = Delay_Resp = 646, all exactly paired.

Scenario	Status	Windows	Capture
baseline	ok	2,669	netem_baseline.pcap — 2,670 packets
pdv (delay variation)	ok	2,589	netem_pdv.pcap — 2,590 packets
loss	ok	392	netem_loss.pcap — 399 packets
holdover	ok	15	netem_holdover.pcap — 17 packets
Total		5,665	matches the figure reported in the deck

4 Reported figures traced to source files

Reported figure	Source file	Recorded value
Live decision latency 0.081 s mean	results/live/live_loop_10min_corrected.summary.json	latency_mean_s = 0.08050435
Live decision latency 0.404 s max	results/live/live_loop_10min_corrected.summary.json	latency_max_s = 0.40352196
10-minute run duration	results/live/live_loop_10min_corrected.summary.json	duration_s = 600.338
5,665 telemetry windows	results/tier2/netem/netem_run_status.csv	2669 + 2589 + 392 + 15
Real-data evaluation split	results/tier2/real_calibration/REAL_CALIBRATION_REPORT.md	379 benign, 7,555 attack windows

5 One correction required before submission

Two of the five “independent” sessions are the same capture.

announce_session_1.pcap and announce_session_2.pcap are byte-identical (MD5 5c6fd791a0ccf22a0d6cfe3031443168). They originate from two upstream files — 15min_announce_attack.pcap and 2024-10-06-announce_attack_Udata.pcap — which are themselves byte-identical to each other. The duplication exists in the published upstream dataset, not in this project's processing. The two label files differ, so the same traffic has been annotated twice.

The corpus therefore contains four distinct captures, not five. Any statement of “five independent sessions” should be corrected to four.

No data leakage resulted. This was checked directly rather than assumed. scripts/calibrate_real.py splits by capture_id using GroupShuffleSplit, which would treat the duplicates as two independent groups and could in principle place identical traffic on both sides of a split. In the recorded run it did not: REAL_CALIBRATION_REPORT.md shows train = announce_session_1, announce_session_2, announce_session_3 and test = sync_followup_session,

sync_singlestep_session. Both duplicates fall in **training**. The reported metrics are unaffected; the practical cost is redundant training data, not an inflated score.

Recommended remedy. Two further genuinely distinct announce captures are present in the upstream repository and currently unused: 2024-10-16-announce_attack.pcap (21.8 MB, the largest available) and prod_successful_announce_attack_ptp.pcap (1.1 MB). Ingesting either restores a true count of five independent sessions. Separately, calibrate_real.py should deduplicate by file checksum rather than by capture_id, so an upstream duplicate can never straddle a split in a future run with a different seed.

6 Independent reproduction of these checks

Every claim above can be re-derived by a third party. From the repository root:

Check	Command
Upstream origin and commit	<code>git -C data/external/s-plane_security_repo remote -v && git -C ... log -1</code>
Capture duplication	<code>md5sum data/external/timesafe_multi_raw/*.pcap</code>
Capture metadata and dates	<code>capinfos data/external/timesafe_multi_raw/*.pcap</code>
Switch identity (LLDP)	<code>tshark -r announce_session_3.pcap -Y lldp -v</code>
PTP profile and message census	<code>tshark -r announce_session_3.pcap -Y ptp -T fields -e ptp.v2.domainnumber</code>
Full result regeneration	<code>python scripts/run_all.py && python scripts/run_tier2.py</code>
Repository self-audit	<code>python scripts/audit_state.py</code>

7 Statement on synthetic data

The project also uses a physics-based simulator. This is **declared synthetic** throughout the project's own documentation and is used for method development and for leave-one-attack-out generalisation testing. It produces no stored files: the directory data/ contains only external/, so nothing synthetic is held on disk alongside the real captures. Simulator-derived percentages must be labelled as such and must not be combined with, or presented as, measured results. The reported real-data figures (99.96% / 99.66% / 99.66% protection, and the 4.49% → 2.37% false-alarm reduction) derive exclusively from the TIMESAFE captures documented in Section 2.

Verification performed 17 August 2026 against the repository working tree at C:\Users\Admin\Documents\AI-Native Self-Healing O-RAN Network using a Digital Twin\01_CURRENT_SPlane_SelfHealing\oran_splane_selfhealing. Checks covered file-format integrity, cryptographic checksums, capture timestamps, link-layer and protocol structure, embedded device identification, and row-level correspondence between captures and derived datasets. Not covered: the correctness of the upstream authors' own labelling, and any claim about hardware the project has not tested (see the project's stated Tier-3 limitations).